AI Hedge Fund Operating Manual

Version 1.0 - AI and Semiconductor Investment Research Platform

Prepared for personal research, paper trading, and disciplined investment decision support.
Research date: June 29, 2026

Important boundary: this manual is not legal, tax, or financial advice. It is a system-design manual for a personal decision-support platform. Do not market it as an investment adviser, do not manage outside money with it, and do not let it auto-trade real capital until it has survived extended paper trading, stress testing, and risk review.

Executive Summary

The objective is to build a boutique AI investment research platform focused first on AI and semiconductor equities. The system should behave like a small institutional investment committee: specialists gather evidence, debate, identify missing information, measure their own performance, and produce one clear daily output. It should not start as a fully autonomous trading bot. The correct first version is a recommendation engine with paper trading, logs, risk controls, and human approval.

The manual is designed around four non-negotiable ideas:

- Data quality beats model size. A powerful AI reading bad or delayed data is just a confident intern with a chainsaw.
- Risk management has veto power. If the trade cannot be sized, stopped, and explained, it is not a trade.
- Every recommendation must be logged and scored. The platform improves by evidence, not vibes.
- Automation is earned. Start with research reports, then paper trading, then small live positions, and only later consider limited auto-execution.

Operating Model

Mission

Create a repeatable daily research process that identifies high-probability opportunities in AI, semiconductor, data-center, infrastructure, and related technology equities while avoiding catastrophic portfolio mistakes.

What the system should produce

- A pre-market market regime assessment: risk-on, risk-off, neutral, or event-driven.
- A ranked watchlist of trade candidates with entry zones, invalidation levels, targets, and confidence scores.
- A no-trade explanation when the edge is weak.
- A portfolio exposure report: sector exposure, concentration, correlation, cash, and maximum loss if stops trigger.
- A post-market scorecard comparing recommendations against outcomes.

- A weekly research director memo identifying missing data, weak agents, overfit strategies, and next improvements.

What the system must not do at first

- It must not place live trades without human approval.
- It must not make recommendations without sources and confidence levels.
- It must not chase every pre-market headline.
- It must not change its own prompts, strategy weights, or risk limits without review.
- It must not promise daily profit. That is how people build expensive slot machines.

Agent Organization Chart

Agent	Primary job	Authority
Chief Investment Officer (CIO)	Converts all specialist reports into one final daily plan. Resolves conflicts and assigns confidence.	Final recommendation authority. Cannot override Risk Manager veto without human approval.
Macro Economist	Rates the broad market environment: rates, inflation, Fed policy, yields, dollar, volatility, liquidity, economic calendar.	Can downgrade risk appetite and reduce position size.
AI Industry Analyst	Tracks AI adoption, hyperscaler capex, model releases, enterprise AI spend, data-center demand.	Can upgrade/downgrade AI demand thesis.
Semiconductor Industry Analyst	Tracks foundries, GPUs, memory, HBM, advanced packaging, lithography, export controls, supply-chain pressure.	Can flag industry-specific catalysts and bottlenecks.
Fundamental Analyst	Evaluates financial statements, margins, cash flow, valuation, earnings quality, revisions, management commentary.	Can reject overvalued or deteriorating businesses for long-biased trades.
Technical Analyst	Maps trend, support/resistance, VWAP, volume, RSI/MACD, moving averages, gaps, and trade triggers.	Can define entry/exit mechanics; cannot create thesis alone.
Options & Flow Analyst	Reads options activity, implied volatility, skew, gamma, volume/open interest, analyst action, unusual institutional behavior.	Can add short-term timing signal; cannot override risk.
Quantitative Researcher	Backtests setups, calculates historical expectancy, measures signal value and false positives.	Can reject setups without measurable edge.
Risk Manager	Controls position sizing, drawdown limits, concentration, liquidity, correlation, stop-loss policy.	Veto authority.
Portfolio Manager	Balances long-term holdings, cash, swing trades, tax awareness, rebalancing, and benchmark	Controls allocation recommendations.

	comparison.	
Behavioral / Devil's Advocate	Looks for FOMO, confirmation bias, crowded trades, weak evidence, and why the thesis could fail.	Can force CIO to include a countercase.
Research Director / Meta Agent	Audits system performance, missing data, bad prompts, weak agents, and improvement proposals.	Can recommend process changes; cannot automatically implement them.

Agent Playbooks

Chief Investment Officer

Mission: The CIO is not another analyst. It is the synthesizer. It must compare all specialist opinions, identify disagreement, and produce a final plan that a human can act on or reject.

- Read every agent report.
- Identify agreements, conflicts, and missing information.
- Produce one daily output: Market Bias, Watchlist, Trade Plan, Risk, Reasons to Avoid, Confidence.
- Never hide disagreement. If agents conflict, show it plainly.
- Prefer no trade over low-quality trade.

Hard rule: Final output must include: ticker, thesis, evidence, entry trigger, stop, target, position size, time horizon, confidence, counterargument, and review date.

Macro Economist

Mission: This agent decides whether the market backdrop supports risk-taking. AI and semiconductor stocks are long-duration, high-expectation assets; rates, liquidity, and volatility matter.

- Track futures, Treasury yields, dollar index, VIX, oil, credit spreads, Fed speeches, CPI/PCE, jobs data, GDP, ISM, and liquidity.
- Classify the day as risk-on, risk-off, neutral, or event-driven.
- Flag major macro events before open.
- Compare today with similar historical regimes.

Hard rule: Never say bullish simply because futures are green. Explain why macro supports or fights the trade.

AI Industry Analyst

Mission: This agent studies whether the AI demand story is strengthening or weakening.

- Track hyperscaler capex, AI model announcements, enterprise adoption, data-center construction, power constraints, cloud revenue, and AI infrastructure order trends.
- Separate hype from actual spend.
- Flag second-order beneficiaries: networking, memory, power, cooling, data-center REITs, software infrastructure.

Hard rule: Every claim must connect to a business driver: revenue, margins, backlog, pricing power, or competitive position.

Semiconductor Industry Analyst

Mission: This agent understands the semiconductor supply chain rather than just ticker symbols.

- Track GPUs, CPUs, HBM, memory cycles, foundry utilization, wafer pricing, advanced packaging, EUV demand, China export controls, Taiwan risk, and equipment lead times.
- Map who benefits from demand and who is capacity constrained.
- Identify bottlenecks before they show up in earnings.

Hard rule: Must explain where the company sits in the stack: design, EDA, foundry, equipment, memory, packaging, networking, or end-market.

Fundamental Analyst

Mission: This agent determines whether the business and valuation justify the trade or investment.

- Review revenue growth, gross margin, operating margin, FCF, capex, debt, valuation multiples, revisions, guidance, and earnings quality.
- Compare management language across quarters.
- Flag accounting oddities and one-time boosts.
- Separate good companies from good stocks at today's price.

Hard rule: For day trades it provides context; for swing/long-term trades it carries more weight.

Technical Analyst

Mission: This agent handles timing only.

- Mark prior high/low, pre-market high/low, VWAP, anchored VWAP, moving averages, volume profile, gap levels, RSI/MACD, ATR, and relative strength.
- Define exact triggers: breakout, pullback, reclaim, failed breakdown, or no-trade.
- Provide invalidation levels and targets.

Hard rule: The technical agent cannot create a trade without a thesis and risk approval.

Options & Flow Analyst

Mission: This agent finds short-term positioning clues.

- Track unusual options volume, IV changes, skew, open interest, put/call ratios, gamma exposure, large premium trades, analyst actions, and institutional signals.
- Distinguish aggressive buying from hedging where possible.
- Flag elevated IV before earnings.

Hard rule: Options flow is a clue, not proof. It must be validated against price and news.

Quantitative Researcher

Mission: This agent is the evidence cop.

- Backtest rules exactly as traded.

- Avoid look-ahead bias and survivorship bias.
- Measure win rate, average win/loss, expectancy, max drawdown, Sharpe, profit factor, and sample size.
- Track confidence calibration.

Hard rule: If a strategy has not been tested, label it untested. No exceptions.

Risk Manager

Mission: This agent exists to keep the account alive.

- Set max daily loss, max weekly loss, max position size, max sector concentration, max open risk, and stop discipline.
- Calculate position sizing from entry to stop.
- Flag correlated bets across NVDA/AMD/AVGO/SMH/QQQ.
- Veto trades during high-risk events or weak liquidity.

Hard rule: Risk Manager has veto power. The system should be annoying about this on purpose.

Portfolio Manager

Mission: This agent manages the whole account, not one shiny ticker.

- Compare active trades against benchmarks such as QQQ, SMH, SOXX, and cash.
- Track core positions vs swing trades.
- Avoid accidental overexposure to one theme.
- Consider tax drag and wash-sale issues for taxable accounts.

Hard rule: The right answer can be: buy the ETF, not the single stock.

Behavioral / Devil's Advocate

Mission: This agent tries to save you from yourself.

- Ask why the trade could fail.
- Identify FOMO, revenge trading, anchoring, overconfidence, recency bias, and confirmation bias.
- Force a counter-thesis into the final report.
- Flag crowded narratives.

Hard rule: The more exciting the trade feels, the harder this agent should push back.

Research Director / Meta Agent

Mission: This agent improves the machine.

- Score each agent over time.
- Identify missing data.
- Recommend new data sources.
- Detect prompt drift.
- Identify which signals are not adding value.
- Suggest experiments for backtesting.

Hard rule: It recommends changes; humans approve them after evidence.

Data Strategy

Good data means data that is timely, accurate, structured, licensed for your use, historically deep enough to test, and specific enough to answer the question. For this project, data quality matters more than GPU power. The platform should start with low-cost sources, prove whether the process creates value, then upgrade selectively.

Data tiers

Tier	Approximate purpose	Examples	When to use
Free / starter	Learning, prototyping, nightly reports, historical research	SEC EDGAR APIs, company IR pages, FRED/economic releases, delayed quotes, free news feeds	Use immediately. Good enough for v1 research and paper trading.
Low-cost developer	More reliable daily workflow with structured fundamentals and market data	FMP, Finnhub, Massive/Polygon-style equities feeds, Nasdaq Data Link datasets	Use when the prototype works and rate limits become annoying.
Trading-grade	Intraday monitoring, options, real-time signals, lower latency	IBKR market subscriptions, Alpaca data, Intrinio options/equities, Benzinga news/ratings	Use after the daily report process is useful and you are paper trading seriously.
Professional / alternative	Differentiated edge from supply chain, capex, hiring, shipping, satellite, power, semiconductor-specific data	Premium Nasdaq Data Link/Sharadar, specialty options feeds, semiconductor research, industry datasets	Use only after you know what missing data would actually improve decisions.

Recommended v1 data stack

Need	Starter source	Upgrade path	Notes
SEC filings and XBRL facts	SEC EDGAR APIs	SEC API / commercial EDGAR parser	Free official EDGAR is excellent but requires normalization.
Fundamentals	FMP or Finnhub	Intrinio, Nasdaq Data Link / Sharadar	Structured financials reduce parsing errors.
Daily/historical prices	Free/delayed API	Massive/Polygon, Intrinio, broker feed	Daily and 1-minute bars are enough for early paper trading.
Broker/paper trading	Alpaca paper or IBKR paper	IBKR live with strict approvals	Paper trading first. No exceptions.
Options data	Delayed options or broker data	Intrinio OptionsEdge / OPRA-grade feed	Options data gets expensive and messy quickly.
News and analyst actions	Free sources + company press releases	Benzinga APIs, Dow Jones/Bloomberg/Refinitiv if budget allows	Speed matters only if you trade intraday.
AI/semi industry	Company IR, earnings transcripts, trade press	Specialized industry research	For edge, this may matter more than chart data.

Data quality checklist

- Timestamp: When was this data created, published, and ingested?
- Latency: Is it real time, 15-minute delayed, end-of-day, or manually updated?
- Coverage: Does it cover all symbols, only US equities, options, ETFs, pre-market, after-hours?
- Point-in-time integrity: Would the strategy have known this data at the time, or is it revised later?
- Corporate actions: Are splits, dividends, ticker changes, and mergers handled correctly?
- Survivorship bias: Does the historical universe include delisted losers or only today's winners?
- Licensing: Are you allowed to store, display, redistribute, or use it in an app?
- Failure mode: What happens if the API returns stale, missing, or incorrect data?

Technology Architecture

The right v1 architecture is boring. Boring is good. Boring means it can be debugged at 6:15 AM when the market is about to open.

Recommended stack

Layer	Recommended v1	Why
Language	Python	Best ecosystem for finance, data, APIs, backtesting, and AI orchestration.
Agent orchestration	OpenAI Agents SDK or LangGraph	Both support multi-agent workflows; keep orchestration explicit and logged.
LLM access	Cloud APIs first	Avoid buying GPUs before proving value.
Structured database	PostgreSQL	Reliable storage for prices, recommendations, trades, agent reports, and metrics.
Vector memory	pgvector or dedicated vector DB	Store transcripts, filings, prior reports, and semantic retrieval.
Workflow scheduling	Prefect or cron initially	Run morning jobs, retries, status checks, and alerts.
Dashboard	Streamlit first	Fastest way to build a working research terminal.
Broker integration	Alpaca paper or IBKR paper	Paper trading and later controlled execution.
Version control	GitHub private repo	Every prompt, strategy, and risk rule should be versioned.

High-level system flow

1. Scheduler starts pre-market workflow.
2. Data ingestion pulls prices, news, macro, filings, earnings, options, and portfolio data.
3. Raw data is written to staging tables with timestamps and source IDs.
4. Normalization layer cleans symbols, corporate actions, time zones, missing fields, and duplicates.
5. Each agent receives only the data it needs plus shared context.

6. Each agent outputs structured JSON plus human-readable commentary.
7. CIO reads all outputs and produces one final plan.
8. Risk Manager validates position sizes and can veto.
9. Human reviews the plan and chooses approve, reject, or paper trade.
10. Post-market workflow scores outcomes and updates performance records.

Reference database schema

```
tables:

  market_bars(symbol, timestamp, timeframe, open, high, low, close, volume, source)

  news_items(id, timestamp, source, symbols, headline, body, url, sentiment, relevance_score)

  filings(id, symbol, form_type, filed_at, period, url, parsed_text, xbrl_json)

  fundamentals(symbol, period, revenue, gross_margin, op_margin, fcf, debt, cash, source)

  options_snapshot(symbol, timestamp, expiration, strike, type, bid, ask, volume, open_interest,
iv, delta, gamma)

  agent_reports(id, run_id, agent_name, timestamp, input_hash, output_json, confidence, citations)

  recommendations(id, run_id, symbol, action, entry, stop, target, size, confidence, thesis,
countercase)

  paper_trades(id, recommendation_id, symbol, entry_time, entry_price, exit_time, exit_price, pnl,
notes)

  performance_metrics(date, strategy, win_rate, expectancy, max_drawdown, sharpe,
benchmark_return)

  prompt_versions(agent_name, version, prompt_text, created_at, notes)
```

Daily Operating Procedures

Pre-market schedule

Time ET	Workflow	Output
5:00 AM	Ingest overnight news, futures, macro calendar, major global market movement.	Raw market regime packet.
5:30 AM	Update watchlist, prices, gaps, earnings calendar, analyst actions, filings.	Ticker candidate list.
6:00 AM	Run specialist agents independently.	Agent reports.
6:45 AM	Run debate pass: agents critique conflicting assumptions.	Conflict memo.
7:15 AM	CIO creates draft plan.	Draft pre-market report.
7:30 AM	Risk Manager validates exposure and sizing.	Approved/rejected trade plan.
8:00 AM	Human review.	Final plan: trade, watch, or skip.
9:30 AM - close	Monitor only approved triggers.	Alerts and paper trades.

4:15 PM	Score plan against actual outcome.	Post-market review.

Pre-market final report template

```
AI Investment Committee - Pre-Market Report

Date:

Market Regime: Risk-on / Risk-off / Neutral / Event-driven

Primary Macro Drivers:

AI/Semiconductor Sector Read:

Top Watchlist:

  1. Symbol - Thesis - Entry trigger - Stop - Target - Size - Confidence

  2. Symbol - Thesis - Entry trigger - Stop - Target - Size - Confidence

Rejected Trades:

  Symbol - Why rejected

Risk Summary:

  Current exposure:

  Max new risk:

  Correlated positions:

Countercase:

Missing Information:

Human Decision: Approve / Paper Trade / Reject
```

Trade recommendation standard

- A recommendation must have an entry trigger, not just a bullish opinion.
- A recommendation must have an invalidation point before entry.
- A recommendation must state maximum dollar loss.
- A recommendation must name the catalyst or setup.
- A recommendation must include at least one reason not to trade it.
- A recommendation must be scored after the trade window closes.

Risk Management Policy

This is the spine of the whole system. Most retail trading systems fail because they focus on being right. Professional systems focus on surviving being wrong.

Suggested starting limits for paper trading

Rule	Starting limit	Purpose
Max risk per trade	0.25% to 0.50% of account	Keeps single-trade losses

		irrelevant.
Max daily loss	1.0% of account	Prevents revenge trading.
Max weekly loss	2.5% of account	Forces cooling-off period.
Max AI/semi concentration	40% to 60% depending on long-term plan	Prevents theme overexposure.
Max single-stock exposure	10% to 15% for swing/investment; less for day trades	Avoids one ticker dominating account.
Max correlated open trades	2 to 3 chip/AI names at once	NVDA, AMD, AVGO, SMH, QQQ often move together.
Earnings rule	No short-dated options through earnings unless explicitly approved	IV crush and gaps are account killers.
News shock rule	No new trade within first minutes after major Fed/CPI/jobs release	Spreads and whipsaw risk spike.

Position sizing formula

Position size should be based on risk, not confidence. If entry is $100, stop is $96, and allowed loss is $200, then the position is 50 shares. Simple. Boring. Effective.

```
risk_per_share = entry_price - stop_price

shares = max_dollar_risk / risk_per_share

position_value = shares * entry_price
```

Risk Manager veto triggers

- No reliable data for the symbol or setup.
- Entry, stop, or target missing.
- Reward-to-risk below minimum threshold.
- Total correlated exposure too high.
- Major event risk within the trade window.
- System confidence is high but historical sample size is weak.
- The trade is based only on news sentiment without price confirmation.
- The user has hit daily/weekly loss limits.

Backtesting and Validation

Backtesting is where pretty ideas go to die. That is good. Kill weak ideas before they touch real money.

Backtest requirements

- Use point-in-time data where possible.
- Include transaction costs and realistic slippage.
- Avoid look-ahead bias: the system cannot use information unavailable at the time.
- Avoid survivorship bias: include delisted symbols where the strategy depends on universe selection.
- Separate in-sample design from out-of-sample validation.
- Measure performance by regime: bull, bear, high-rate, low-rate, high-volatility, earnings season.
- Compare against dumb benchmarks: QQQ, SMH, SOXX, and buy-and-hold cash/ETF alternatives.

Metrics that matter

Metric	Meaning	Why it matters
Expectancy	Average expected profit/loss per trade	More important than win rate.
Profit factor	Gross wins divided by gross losses	Shows whether winners meaningfully exceed losers.
Max drawdown	Largest peak-to-trough decline	Tells you how ugly it can get.
Sharpe / Sortino	Risk-adjusted returns	Useful but can be gamed by smooth backtests.
Hit rate	Percent profitable trades	Can be misleading if losers are huge.
Average win/loss	Magnitude of winners and losers	Tells if risk/reward is real.
Confidence calibration	Whether 80% confidence is actually right about 80% of the time	Prevents false certainty.
Benchmark excess return	Return above QQQ/SMH/SOXX after costs	If it cannot beat the ETF, buy the ETF.

Agent Communication Protocol

Agents should not ramble at each other. They should exchange structured data. Natural language is useful for humans; JSON is useful for systems.

Standard agent output schema

```
{
  "agent": "Technical Analyst",
  "run_id": "2026-06-29-premarket",
  "symbol": "NVDA",
  "stance": "bullish | bearish | neutral | no_trade",
  "confidence": 0.72,
  "time_horizon": "intraday | swing | investment",
  "key_evidence": ["...", "..."],
  "levels": {"entry": null, "stop": null, "target_1": null, "target_2": null},
  "risks": ["..."],
  "missing_information": ["..."],
  "citations": [{"source": "...", "url": "...", "timestamp": "..."}]
}
```

Debate protocol

11. Each agent submits its independent report before seeing the others.

12. CIO shares summarized findings with all agents.
13. Each agent may challenge only claims related to its domain.
14. Devil's Advocate writes the strongest countercase.
15. Quant Agent checks whether the proposed setup has historical support.
16. Risk Manager validates whether the trade can be sized safely.
17. CIO produces the final decision and explains disagreements.

Prompt Playbooks

Universal agent constitution

```
You are a specialist agent inside a personal investment research platform.

Rules:

1. Do not invent facts. If data is missing, say exactly what is missing.

2. Separate fact, inference, and opinion.

3. State confidence as a number from 0.00 to 1.00.

4. Include reasons your conclusion may be wrong.

5. Prefer no-trade when the evidence is weak.

6. Cite every non-obvious external fact with source and timestamp.

7. Never recommend position size without a stop/invalidation level.

8. Never optimize for excitement. Optimize for repeatable expectancy and survival.
```

CIO prompt skeleton

```
Given the specialist reports, create one pre-market investment committee output.

You must:

- Identify the market regime.

- Rank opportunities by expected quality, not excitement.

- Reject weak trades explicitly.

- Include entry trigger, stop, target, position size, confidence, and countercase.

- Show agent disagreements.

- Respect Risk Manager vetoes.

- End with one of: TRADE APPROVED, PAPER TRADE ONLY, WATCH ONLY, or NO TRADE.
```

Research Director prompt skeleton

```
Audit the system after today's run.

Report:
```

```
- Which agents had low confidence and why.

- What missing data would have most improved the decision.

- Whether any agent made unsupported claims.

- Which recommendations should be backtested.

- Whether any prompt or data pipeline should be changed.

- Do not implement changes. Propose them for review.
```

Build Roadmap

Phase	Goal	Deliverables	Exit criteria
0 - Design	Define scope and guardrails	Manual, watchlist, agent specs, risk rules	Manual approved; no code yet.
1 - Single report	Generate daily pre-market report manually/semi-automated	Data ingestion, basic agents, CIO summary	30 consecutive reports produced.
2 - Multi-agent debate	Specialists produce independent reports and CIO reconciles	Structured agent outputs, conflict memo	Reports are consistent and useful.
3 - Paper trading	Convert plans into tracked paper trades	Paper broker integration, trade log, dashboard	60-90 days of paper results.
4 - Backtesting	Validate repeatable setups	Backtest engine, metrics, benchmark comparison	Positive expectancy after costs in out-of-sample tests.
5 - Small live pilot	Human-approved small trades only	Execution checklist, kill switch, limits	Live results match paper assumptions.
6 - Controlled automation	Limited auto-execution for narrow setups	Hardcoded risk limits, monitoring, fail-safes	Only if prior phases justify it.

Budget Options

Budget	Monthly spend	What it buys	Best use
Lean prototype	$0 - $50	Free SEC data, free/delayed market data, manual news collection, cloud LLM pay-as-you-go	Learning and first reports.
Serious personal research	$100 - $300	Structured fundamentals, better historical data, basic market/news APIs, cloud LLM usage	Best starting target.
Advanced paper trading	$300 - $1,000	Real-time equities, delayed/real-time	When v1 proves useful.

		options, analyst/news APIs, better compute/cloud hosting	
Professional-grade	$1,000+	Premium options, alternative datasets, institutional news, specialized semi research	Only after measured ROI.

Practical recommendation: start in the $100-$300/month range before spending thousands on local GPUs. Spend on data, logging, and validation first.

Compliance, Legal, and Ethical Guardrails

For personal use, the platform can be a powerful research assistant. The moment you provide individualized advice to others, manage outside money, market performance, or charge fees, the legal picture changes. Get securities counsel before crossing that line.

Core guardrails

- Personal use only unless reviewed by qualified securities counsel.
- Do not call it a hedge fund if it is not a legally formed and compliant investment vehicle.
- Do not advertise AI capabilities or performance that are not real and documented.
- Keep complete logs of data, prompts, outputs, human decisions, and trade actions.
- Use broker paper trading before live trading.
- Respect market-data licensing restrictions, especially for redistribution/display.
- Create a kill switch for execution systems.
- Separate research from execution; require human approval until the process is proven.

Regulatory themes to respect

SEC and FINRA materials emphasize supervision, controls, truthful AI representations, fiduciary duties for advisers, and the risks of automated investment advice and algorithmic trading. Even for personal use, those themes are useful design principles: document what the system does, validate it, supervise it, and do not overstate it.

Security and Operational Controls

- Store API keys in a secrets manager or environment variables, never in prompts or code commits.
- Use read-only data credentials wherever possible.
- Separate paper and live broker keys.
- Require manual approval for live order submission.
- Log every external API response used in a recommendation.
- Create stale-data checks before market open.
- Add circuit breakers for API failure, abnormal spread, abnormal volatility, and daily loss limits.
- Back up PostgreSQL and prompt versions.
- Use Git branches and pull requests for prompt/risk-rule changes.
- Run monthly disaster recovery test: can the system rebuild from backups?

Performance Review Cadence

Daily

- Was the market regime correct?
- Did approved triggers occur?
- Were entries realistic?
- Was slippage acceptable?
- Did the Risk Manager prevent bad trades?
- What was missed?

Weekly

- Agent accuracy by domain.
- Best and worst recommendations.
- Missed opportunities.
- False positives.
- Data failures.
- Prompt issues.
- Benchmark comparison.

Monthly

- Strategy expectancy.
- Drawdown profile.
- Confidence calibration.
- Whether any agent should gain/lose weight.
- Whether any data subscription earned its keep.
- Whether to expand, pause, or simplify.

Initial Watchlist Universe

The initial universe should be focused but not tunnel-visioned. A tight list makes the system easier to validate.

Category	Example tickers	Why included
AI compute leaders	NVDA, AMD, AVGO	Core AI accelerator and custom silicon exposure.
Foundry / manufacturing	TSM	Critical manufacturing bottleneck and geopolitical risk center.
Equipment / lithography	ASML, AMAT, LRCX, KLAC	Capex cycle and leading-edge fabrication exposure.
Memory / HBM	MU, Samsung/foreign where accessible, SK Hynix/foreign where accessible	HBM and memory cycle exposure.
Networking / data center	ANET, MRVL, AVGO	AI cluster networking and custom ASIC exposure.
Cloud / hyperscalers	MSFT, GOOGL, AMZN, META	AI capex demand source and monetization check.
ETFs / benchmarks	QQQ, SMH, SOXX	Benchmarks and lower-risk alternatives.

First 30-Day Action Plan

18. Create a private GitHub repository.
19. Create PostgreSQL database with the schema in this manual.
20. Pick 10-20 tickers for initial universe.
21. Set up free SEC EDGAR ingestion and one market-data API.
22. Build a Streamlit dashboard showing watchlist, news, prices, and agent reports.
23. Implement three agents first: Macro, Technical, Risk.
24. Add CIO summary after the first agents work.
25. Generate daily reports for 30 market days without trading.
26. Add paper trading only after the reports are stable.
27. Review whether the outputs are better than simply buying SMH/QQQ.

Source Notes and Current References

These sources were reviewed on June 29, 2026 for current platform design. URLs and offerings change, so re-check vendor pricing and licensing before subscribing.

28. **OpenAI Agents SDK documentation:** https://developers.openai.com/api/docs/guides/agents - Used for agent orchestration concepts and OpenAI recommendation to use Agents SDK when the application owns orchestration, tools, approvals, and state.
29. **OpenAI Agents Python SDK:** https://openai.github.io/openai-agents-python/agents/ - Used for SDK concepts including agents, runner, tools, guardrails, handoffs, and sessions.
30. **Interactive Brokers Web API Trading:** https://www.interactivebrokers.com/campus/ibkr-api-page/web-api-trading/ - Used for broker API and account/market data/trading integration references.
31. **Interactive Brokers Paper Trading Account:** https://www.interactivebrokers.com/campus/glossary-terms/paper-trading-account/ - Used for paper trading availability through IBKR APIs.
32. **Alpaca developer/API platform:** https://alpaca.markets/ - Used for developer-first API and algorithmic trading/paper trading platform reference.
33. **SEC EDGAR APIs:** https://www.sec.gov/search-filings/edgar-application-programming-interfaces - Used for official free SEC submissions and XBRL/company facts APIs.
34. **SEC Developer Resources:** https://www.sec.gov/about/developer-resources - Used for official confirmation of RESTful APIs on data.sec.gov.
35. **FINRA Algorithmic Trading topic page:** https://www.finra.org/rules-guidance/key-topics/algorithmic-trading - Used for algorithmic trading supervision/control theme.
36. **FINRA Regulatory Notice 15-09:** https://www.finra.org/rules-guidance/notices/15-09 - Used for effective supervision and control practices for algorithmic strategies.
37. **SEC speech on AI and investment management:** https://www.sec.gov/newsroom/speeches-statements/daly-020326-artificial-intelligence-future-investment-management - Used for current SEC framing of AI risks/opportunities in investment management.
38. **SEC/Investor.gov Robo-Adviser bulletin:** https://www.investor.gov/introduction-investing/general-resources/news-alerts/alerts-bulletins/investor-bulletins-45 - Used for automated advice concepts and investor-risk framing.

39. **Nasdaq Data Link API:** https://www.nasdaq.com/solutions/data/nasdaq-data-link/api - Used for data API and market-data delivery reference.
40. **Nasdaq Data Link documentation:** https://docs.data.nasdaq.com/docs/getting-started - Used for free/premium datasets, streaming, REST, and table APIs.
41. **FMP pricing and docs:** https://site.financialmodelingprep.com/developer/docs/pricing - Used for structured fundamentals and stock market API reference.
42. **Finnhub pricing and docs:** https://finnhub.io/pricing - Used for free and paid market/fundamental data API reference.
43. **Intrinio options/equities data:** https://intrinio.com/options - Used for options-data feed concepts and OPRA/licensing complexity.
44. **Intrinio pricing:** https://intrinio.com/pricing - Used for options, equities, fundamentals, and delayed/real-time data categories.
45. **Benzinga APIs:** https://www.benzinga.com/apis/ - Used for market-moving news, analyst ratings, and financial data API categories.
46. **Massive/Polygon pricing:** https://massive.com/pricing - Used for equities market-data tier examples including real-time/full-market plans.
47. **LangGraph multi-agent workflows:** https://www.langchain.com/blog/langgraph-multi-agent-workflows - Used for multi-agent workflow design reference.
48. **LangChain multi-agent docs:** https://docs.langchain.com/oss/python/langchain/multi-agent - Used for specialist component coordination concepts.
49. **PostgreSQL official site:** https://www.postgresql.org/ - Used for relational database foundation reference.
50. **Prefect documentation:** https://docs.prefect.io/v3/get-started - Used for workflow orchestration, scheduling, retries, and monitoring concepts.
51. **Streamlit documentation:** https://docs.streamlit.io/ - Used for dashboard/data app prototyping reference.

Final Build Philosophy

The most valuable version of this platform is not the flashiest. It is the one that gives you fewer, better decisions; tells you when it does not know; refuses bad trades; and improves because it measures itself honestly. Build it like a research firm first. Let it earn the right to become a trading system later.